Crystallographic Closure of Integral Lorentzian Lattices and the Rank at Which It Ends

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2026-08-27

A finite-order integral isometry of any lattice of signature $(3,1)$ has order in $1,2,3,4,6$; for $I_{3,1}$ all five are realized. On signature $(4,1)$ this is no longer true.

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1 · Statement

Lemma 1 (integral timelike axis). Let Λ be an integral lattice of signature $(n,1)$, $g \in O(\Lambda)$ a finite-order isometry. Then there exists an **integral** vector $t \in \Lambda$ with $q(t) < 0$ and $\varepsilon \in \pm 1$ such that $g \cdot t = \varepsilon \cdot t$.

Theorem 1 (reduction). In the notation of Lemma 1, the lattice $L := t^{\perp} \cap \Lambda$ is **positive definite**, has rank n, is invariant under g, and $ord(g) = lcm(ord(g|_L), ord(\varepsilon))$. Hence the set of orders on signature $(n,1)$ **equals** the set of orders on positive definite lattices of rank n.

Theorem 2 ($n = 3$). For **any** integral lattice of signature $(3,1)$: $ord(g) \in 1,2,3,4,6$.

Corollary (sharpness). For $I_{3,1}$ **all five** are realized: $ord(g) : g \in O(I_{3,1}), ord(g) < \infty = 1,2,3,4,6$.

Unimodularity does no work anywhere. It is needed only for the corollary — to name a specific lattice on which all five orders are present. The upper bound rests on **integrality** and **negative index 1**, and on nothing else.

This is the same restriction as in three-dimensional crystallography — but here it is proved for a **Lorentzian** lattice, and the reason it survives is the content of §4. The reason it **ends** at the next rank is the content of §5.

2 · Proof

Lemma 1 — averaging over the finite orbit

(L.1.1) Λ contains an integral timelike vector. Λ spans $\Lambda \otimes \mathbb{R}$, the condition $q < 0$ is **open** and nonempty (signature $(n,1)$), and $\Lambda \otimes \mathbb{Q}$ is dense in $\Lambda \otimes \mathbb{R} \Longrightarrow$ there exists a rational vector with $q < 0$; multiplying by a common denominator gives an integral one. Denote it e.

(L.1.2) Choice of sign. The set $q < 0$ has exactly **two** connected components (a cone and its opposite). An isometry either preserves both or swaps them; for two timelike vectors u, v they lie in the same component if and only if $\langle u, v\rangle < 0$. Hence there exists $\varepsilon \in \pm 1$ such that $h := \varepsilon \cdot g$ **preserves** the component of e. Together with g, the matrix h has finite order; let $m := ord(h)$.

(L.1.3) Averaging. Set

```
t := Σ_{j=0}^{m−1} h^j · e
```

- t is **integral**: h is integral, e is integral;
- t is **timelike and nonzero**: all $h^j \cdot e$ lie in **one** component, and it is a **convex** cone, so the sum stays in it $\implies q(t) < 0$, in particular $t \neq 0$;
- $h \cdot t = t$, because $h^m = 1$ only permutes the summands. Since $h = \varepsilon \cdot g$ and $\varepsilon^2 = 1$, we get $g \cdot t = \varepsilon \cdot t$. ∎

Why averaging specifically. The seemingly more natural route — from the compactness of the generated group to an invariant **real** line, and then to its rationality — has a gap: rationality of a proper *subspace* does not imply rationality of a specific *line* within it. A witness — rotation by $\pi/2$ in the plane (x_1, x_2) on $I_{3,1}$: the kernel $ker(g-1) = \langle e_3, e_4\rangle$ is rational, but contains an invariant timelike line $(0, 0, 1, \sqrt{}2)$, which is not rational. Along with it, the rank-3 section would fall too. Averaging gives an axis that is **integral** from the start, constructively, and without any theory of compact groups.

Theorem 1 — reduction

t is timelike $\implies t^\perp$ is positive definite, so $L = t^\perp \cap \Lambda$ is a positive definite integral lattice. t is integral $\implies t^\perp$ is rational $\implies rankL = n$. From $g \cdot t = \pm t$ it follows that $g(t^\perp) = t^\perp$, hence $g \cdot L = L$.

The sum $L \oplus \mathbb{Z}t$ has **finite index** in Λ, and an isometry that is the identity on a finite-index sublattice is the identity on all of Λ (both span the same $\mathbb{Q}$-space). Therefore $ord(g) = lcm(ord(g|_L), ord(g|_{\mathbb{Z}t}))$, and $g|_{\mathbb{Z}t} = \varepsilon$ has order 1 or 2. ∎

Equality of sets — both inclusions.

($\subseteq$) Let $k := ord(g|_L)$. If $\varepsilon = +1$ **or** k is even, then $ord(g) = lcm(k, ord\varepsilon) = k$, and this order is already carried by $g|_L$ itself on the positive definite L. If $\varepsilon = -1$ **and** k is odd, then $ord(g) = 2k$, and the isometry needed is given by $-g|_L$: it is integral, preserves the same L, and from $(-g|_L)^m = (-1)^m \cdot (g|_L)^m$, with k odd, the identity is first reached at $m = 2k$, i.e. $ord(-g|_L) = 2k = ord(g)$.

($\supseteq$) If L_0 is a positive definite integral lattice of rank n with an isometry of order k, then $L_0 \oplus \langle -1\rangle$ has signature $(n, 1)$ and carries the same isometry $\oplus 1$. ∎

Lemma 2 — definite rank 3: a trace argument

> An integral isometry h of finite order of a **positive definite** lattice of rank 3 has $ord(h) \in 1, 2, 3, 4, 6$.

Over $\mathbb{R}$ such h lies in $O(3)$, hence is conjugate to $R_\theta \oplus (\delta)$, $\delta = \det(h) \in \pm 1$, where R_θ is a rotation of the plane. The trace does not depend on the basis, and in the lattice basis the matrix is **integral**:

```
tr(h) = δ + 2·cos θ ∈ ℤ ⟹ 2·cos θ ∈ ℤ ⟹ cos θ ∈ {0, ±½, ±1}
```

These are exactly the possible orders of the rotation $r := ord(R_\theta) \in 1,2,3,4,6$. Next $ord(h) = lcm(r, ord(\delta))$, and

```
r :  1 2 3 4 6
lcm(2, r):  2 2 6 4 6 -- all in {1,2,3,4,6}
```

so the set remains closed in the case $\delta = -1$ as well. ■

Three different quantities, not one. r is the order of the planar **rotation**; $k = ord(h)$ is the order of the **whole** rank-3 isometry; $ord(g)$ is the order of the original Lorentzian isometry. $lcm(2,\cdot)$ is applied twice for different reasons: inside Lemma 2 (the δ block) and in Theorem 1 (the axis ε).

Zero imports. Here a reference to the classical crystallographic restriction could have stood (Scherrer 1946, Coxeter 1969). It does not: the argument uses only the **integrality of the trace**, and this is three lines. The rest of the classical facts are also derived on the spot: a finite-order integral matrix is semisimple with roots of unity (the minimal polynomial divides $x^m - 1$, which is separable over $\mathbb{Q}$); a finite subgroup of $GL_n(\mathbb{Z})$ preserves a positive definite form (averaging). Scherrer and Coxeter remain a **historical reference, not a leg of the proof**.

Theorem 2 and the corollary

Theorem 1 at $n = 3$ together with Lemma 2 gives $ord(g) \in 1,2,3,4,6$. Sharpness for $I_{3,1}$ — by exhibition; $\eta = \mathrm{diag}(1,1,1,-1)$:

order	integral isometry	why this order
1	I_4	—
2	$-I_4$	—
3	cyclic permutation $x_1 \to x_2 \to x_3 \to x_1$	order of the cycle
4	rotation by $\pi/2$ in the plane (x_1, x_2)	$lcm(4,1)$
6	the same permutation $\oplus(-1)$ on x_4	$lcm(3,2) = 6$

All five are integral, all preserve η. ■

3 · Exact scope — in two tags

The text must not claim more than the probe measured. Hence:

statement	tag	what supports it
Lemma 1, Theorem 1, Lemma 2	**derived**	§2, in full; zero imports
orders $1,2,3,4,6$ are realized on $I_{3,1}$	**derived**	the five explicit matrices above — multiplied by hand

statement	tag	what supports it
orders $5, 8, 10, 12$ do not occur on signature $(3, 1)$	**derived**	Theorem 2 (Lemma 1 + Lemma 2). The parity lemma (§4) gives only a **structural reason** — the mandatory ± 1 block; the prohibition itself is completed by the fact that a **definite rank 3** remains in the orthogonal complement
the same, by an **independent route**	**measured**	enumeration of **all 24 types** of rank 4 and all reachable signatures
Φ_5, Φ_{10} do not occur on **any unimodular** lattice of rank 4	**measured**	square class of det $= 5$ (§4)
Φ_8 acts on $I_{2,2}$	**measured**	explicit matrix of order 8 (§4)
table across ranks 2–8	**measured**	structural count, checked against linear algebra type by type

Machine check against a known answer: enumerating integral isometries gave $\mathbb{Z}^3$ — 48 elements, orders $1, 2, 3, 4, 6$; A_3 — 48, the same; hexagonal — 24, $1, 2, 3, 6$. The union is $1, 2, 3, 4, 6$, **no order 5 at all**. This is a **check**, not a leg: the verdict is carried by the trace argument.

4 · What actually filters — and what does not

The mechanism — three parts, not one

negative index 1 + integrality + rank 4

The roles are separated:

1. **negative index 1** makes the negative block one-dimensional $\Longrightarrow$ a finite-order action on it is only ± 1 $\Longrightarrow$ **the parity lemma**: the isotypic components of different real irreducible constituents are orthogonal for any invariant form, while on a rotation block ($\theta \neq 0, \pi$) invariant symmetric forms correspond to **Hermitian** forms, and hence give an **even** negative index. An odd index $\Longrightarrow$ a ± 1 block **must** be present;

2. **integrality** turns this axis into an **arithmetic** one (Lemma 1) and leaves an integral lattice in the orthogonal complement;

3. **rank 4** turns this lattice into rank 3, where the trace quantizes $2cos\theta$.

Signature alone filters nothing. In the real group $O(3, 1)$ an element of order 5 does exist: rotation by $2\pi/5$ in a spatial plane preserves $\mathrm{diag}(1, 1, 1, -1)$ and has order exactly 5. It is simply **not integral** ($cos(2\pi/5) = (\sqrt{5} - 1)/4$).

"Exactly one timelike direction" is false. There are infinitely many timelike directions ($(0, 0, 0, 1)$, $(1, 0, 0, 2)$, $(0, 1, 1, 2)$ — already three independent ones). The correct formulation is: **the maximal negative definite subspace is one-dimensional**, i.e. $negative index = 1$.

Non-crystallographic axes: three different verdicts, not one

The tempting phrase "Φ_5, Φ_8, Φ_{12} can act on $I_{2,2}$, but not on $I_{3,1}$" merges **three statements of different truth-status**. One invariant distinguishes them.

Square class. An integral invariant form on **any** $\mathbb{Z}$-lattice within the same $\mathbb{Q}$-reduction is a rational point of the same family, scaled by $\det(P)^2$ of the change of basis. Hence the class of det in $\mathbb{Q}*/(\mathbb{Q}*)^2$ **does not depend on the lattice**, and the question "does a unimodular one occur" is settled for all lattices at once — by a single number, without enumeration:

type	det **of the family of invariant forms**	**class**	**unimodular lattice**
Φ_5	$5(4p_0^2+2p_0p_1-p_1^2)^2/16$	**5**	**impossible, none**
Φ_{10}	$5(4p_0^2-2p_0p_1-p_1^2)^2/16$	**5**	**impossible, none**
$\Phi_3+\Phi_4$, $\Phi_4+\Phi_6$	$3p_0^2p_1^2/4$	**3**	**impossible, none**
Φ_8	$(2p_0^2-p_1^2)^2$	1	**exists**, and odd at $(2,2)$
Φ_{12}	$(4p_0^2-3p_1^2)^2/16$	1	exists, but found to be **even**

Hence:

- **order 5** (and 10) does not act on $I_{2,2}$ — and does not act on any unimodular lattice of rank 4;
- **order 8** does act, and precisely on $I_{2,2}$. To avoid relying on the classification of indefinite odd unimodular lattices, the form was diagonalized by an **explicit unimodular change of basis** to $\mathrm{diag}(1,1,-1,-1)$, and the isometry was carried into this basis:

  ```
  M = [ −3 −3 −1 −4 ]
  [ 3 −3 −4 1 ] MᵀηM = η for η = diag(1,1,−1,−1), ord(M) = 8
  [ −4 1 3 −3 ]
  [ 1 4 3 3 ]
  ```

- **order 12**: a unimodular invariant form of signature $(2,2)$ exists, but the one found is **even**, i.e. it is $II_{2,2}$, not $I_{2,2}$. Whether an odd one exists is **open**: parity also depends on the lattice, so a search within a bounded box proves nothing here.

Why this is not a bookkeeping detail. The "safe" formulation ("they admit integral nondegenerate invariant forms of signature (2,2), but not (3,1)") is true — and it blurs precisely this distinction, leaving the reader with the impression that all three cases are the same.

Answer to the natural question "what about quasiperiodic axes?": in this arithmetic there are none, and the reason is named explicitly — not "not found", but **negative index 1 together with integrality in rank 4 leaves no room for them**.

5 · Where the closure ends

Theorem 1 reduces the Lorentzian question of rank $n+1$ to the definite question of rank n. Measured for ranks 2–8; at rank 4 the structural count was checked against linear algebra type by type:

rank	signature	orders
2	$(1,1)$	$1,2$
3	$(2,1)$	$1,2,3,4,6$
4	$(3,1)$	$1,2,3,4,6$
5	$(4,1)$	$1,2,3,4,5,6,8,10,12$
6	$(5,1)$	$1,2,3,4,5,6,8,10,12$
7	$(6,1)$	$1,2,3,4,5,6,7,8,9,10,12,14,15,18,20,24,30$
8	$(7,1)$	same as at rank 7

The closure breaks exactly at signature $(4,1)$, and it breaks substantively: $\Phi_5 \oplus \Phi_1$ has degree 5 and signature $(4,0)+(0,1)=(4,1)$ — the fivefold axis goes through as soon as the definite complement has enough rank to hold it. Next comes a **plateau** (rank 6 adds nothing), and the next break is at $(6,1)$, where order 7 appears, because $\varphi(7)=6$.

An identity, read off from the table and verified for ranks 3–8:

```
{ orders on signature (n−1,1) } = { orders on a definite lattice of rank n−1 }
```

This is Theorem 1 seen from the other side: **for the spectrum of finite orders**, the Lorentzian problem $(n,1)$ adds nothing beyond the definite question of rank n. **The formulation is deliberately narrow:** it is about the **set of orders**, not about the classification of Lorentzian lattices in general. Within these limits, "the crystallographicity of signature $(3,1)$" is not a property of Lorentzian-ness, but a property of **the number 3**: rank 3 turns out to be the **last definite rank** where the spectrum still equals the classical $1,2,3,4,6$.

6 · Limits

- **Multiplicity 1.** This statement and the three-dimensional crystallographic restriction are **not two witnesses**: Theorem 1 **reduces** the former to the latter, a common ancestor $\Longrightarrow$ multiplicity 1. The same holds for the two routes to the prohibition of order 5 — the signature one and the trace one: both come out of **one and the same** integral arithmetic. This is **corroboration**, not independent confirmation.
- **Window (quasiperiodicity) is a separate input, not a consequence.** If a structure admits a window, this is a condition introduced **from outside**; what is measured here is what integral Lorentzian arithmetic allows.
- **No physical reading whatsoever.** The term "timelike" is used **only** in the sense of the sign of the quadratic form; no interpretation of a time coordinate is introduced.
- **Open:** whether order 12 acts on $I_{2,2}$ (§4). A general description of the square class of det for an arbitrary cyclotomic type was not sought here.
- **Zero imports in the chain.** What is telling is **how** the last debt was closed: the first attribution attached to the step of Lemma 2 ("Hermann–Mauguin") turned out to be **the name of a notation mistaken for a theorem**, and was removed; instead of looking for a replacement, the step was derived. It proved cheaper to **prove** than to cite correctly. The same thing happened a second time in §4: the classification of unimodular lattices was replaced by explicit diagonalization.

7 · Relation to preprint-7

what	where
full statement, proof, scope	**here**
a line about the scope, **without a reference here**	P7 §11(d)
a reference **from here to P7** — legitimate	P7 is published, concept DOI 10.5281/*zenodo*.22125370
a reference **from P7 to here** — not yet	waits for this work's address

A forward-reference is not placed: a reference to a nonexistent artifact is broken from day one and is fixed only by a new version. The asymmetry is deliberate and temporary: when this work receives an address, the line at P7 §11(d) will receive a reference **via a new version of P7**, not by editing what is published.

Preprint-7: *Section Criteria for Integral Lorentzian Lattices and the Cost of Realization*, Zenodo [10.5281/zenodo.22125370](https://doi.org/10.5281/zenodo.22125370).

Acknowledgement

This text went through **two rounds of external review**. The first pointed to a gap in the proof of the rationality of the axis, to an incomplete measurement for order 12, and to an overstated formulation about $I_{2,2}$; the second — to the fact that Theorem 1 claimed equality of sets while only one inclusion was proved, and to two overstated formulations in §3 and §5. All remarks were accepted: the proof route was rebuilt, §4 was rewritten, the inclusion $\subseteq$ was added. The errors that remain are the author's.